

Five benchmark datasets

Designed barcode architecture, error regimes and input scale

Dataset	Barcode design (N = variable base)	Errors and scale
R-S Random / substitutions	NNNNNNNNNNNNNNNNNNNN 20 bases; 20 variable positions	Substitutions: 0.5% per base 10,000 truth barcodes; 1 million reads / seed
R-I Random / mixed errors	NNNNNNNNNNNNNNNNNNNN 20 bases; 20 variable positions	Substitutions + insertions + deletions 0.5% each; same scale as R-S
A-S Anchored / substitutions	NNNNNNNN ATGC NNNNNNNN ATCGTTAA 28 bases; 16 variable positions	Substitutions: 0.5% per base 10,000 truth barcodes; 1 million reads / seed
A-I Anchored / mixed errors	NNNNNNNN ATGC NNNNNNNN ATCGTTAA 28 bases; 16 variable positions	Substitutions + insertions + deletions 0.5% each; same scale as A-S
Milos Johnson reference simulation	NNNNNNNNNNNNNNNNNNNN 20 bases; 20 variable positions	Substitutions + sparse length-changing errors 100,000 truth barcodes; 24,996,128 reads

R-S, R-I, A-S, A-I: independent seeds 42, 43, 44; lognormal abundance (sigma = 1.5).

Milos: one fixed input; 1,544,850 observed unique sequences. 1,001 reads differ in length from their parent.

Schematics show the truth designs. Indels change observed lengths; matching length does not exclude an indel history.